

Proyecto segundo corte

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Introducción

En el contexto actual de administración de sistemas y redes, la capacidad de implementar y monitorear infraestructuras virtualizadas es una competencia fundamental. Este proyecto tiene como objetivo principal desarrollar un entorno híbrido de pruebas compuesto por máquinas virtuales (VMs) y contenedores Docker, que permita evaluar el comportamiento de diferentes sistemas operativos en términos de consumo de recursos, exposición de servicios y análisis de tráfico de red.

A través del uso de herramientas de análisis como htop, ps aux, tcpdump, nmap, netstat, así como soluciones de monitoreo como Prometheus y Grafana, se logró implementar una arquitectura. Además, se trabajó con agentes de monitoreo, configuraciones de red personalizadas y la visualización de datos mediante dashboards interactivos. Este documento detalla las fases del proyecto, sus objetivos, resultados obtenidos y conclusiones técnicas, cumpliendo con los criterios del segundo corte académico.

Marco Teórico

Las máquinas virtuales (VMs) permiten la emulación completa de sistemas operativos sobre un mismo hardware físico, ofreciendo entornos aislados y estables. Por otro lado, los contenedores Docker proporcionan una alternativa más ligera al compartir el mismo kernel del sistema anfitrión, permitiendo una rápida implementación y escalabilidad.

Ambas tecnologías tienen ventajas y limitaciones: las VMs consumen más recursos pero ofrecen mejor aislamiento; los contenedores son más rápidos y eficientes en el uso de memoria, pero requieren una configuración adicional para exponer servicios y asegurar persistencia.

Las herramientas de monitoreo como Prometheus, junto a visualizadores como Grafana, permiten centralizar y graficar métricas del sistema, mientras que utilidades como htop, tcpdump, netstat o nmap permiten realizar un diagnóstico detallado de procesos, tráfico de red y servicios activos. Este proyecto aplica dichas herramientas sobre VMs (Rocky, Arch, Manjaro) y contenedores (Fedora, Alpine, Garuda, Debian) para identificar diferencias clave en su operación.

Objetivo General

Implementar una infraestructura híbrida compuesta por máquinas virtuales y contenedores Docker que permita analizar y comparar el comportamiento de diferentes sistemas operativos mediante herramientas de monitoreo, análisis de red y visualización de métricas.

Objetivos Específicos

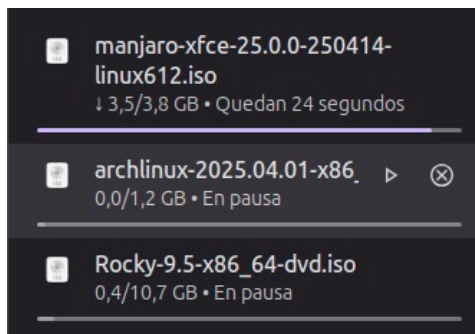
- Instalar y configurar tres máquinas virtuales (Rocky, Arch, Manjaro) sobre QEMU/KVM.
- Crear tres contenedores Docker (Garuda, Debian, Alpine) y un contenedor Fedora Central personalizado con herramientas de análisis.
- Desarrollar subredes independientes para VMs y contenedores, asegurando comunicación cruzada.
- Analizar servicios y procesos mediante herramientas como htop, ps aux, glances, netstat, bpytop y journalctl.
- Capturar tráfico de red utilizando tcpdump, iftop, nethogs y wireshark.
- Visualizar datos en tiempo real usando Prometheus y Grafana.

1. Instalación de Máquinas Virtuales (Primer Criterio)

- **Descripción:**
Instalación de las tres máquinas virtuales Arch Linux, Manjaro Linux y Rocky Linux.

- **Pasos realizados:**

Paso 1:



- Instalar Isos

Paso 2:

```

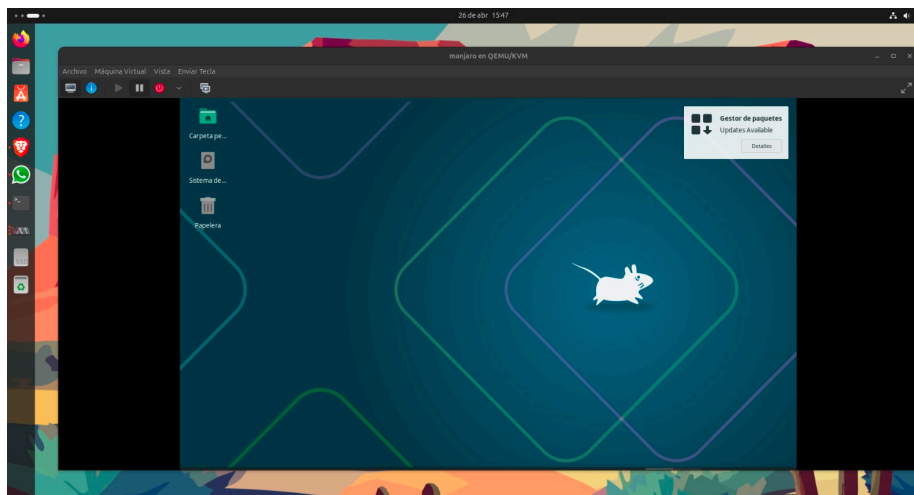
ker@ker-XPS-8910:~$ history
1  sudo apt update -y
2  sudo apt upgrade -y
3  kvm-ok verificar qemu
4  sudo apt install cpu-checker
5  egrep -c '(vxlsvm)' /proc/cpuinfo
6  sudo apt install qemu-kvm virt-manager virtinst libvirt-clients bridge-ut
ils libvirt-daemon-system
7  sudo systemctl enable --now libvirtd
8  sudo systemctl start libvirtd
9  sudo systemctl status libvirtd
10 sudo usermod -aG kvm $USER
11 sudo usermod -aG libvirt $USER
12 sudo virt-manager
13 history
ker@ker-XPS-8910:~$

```

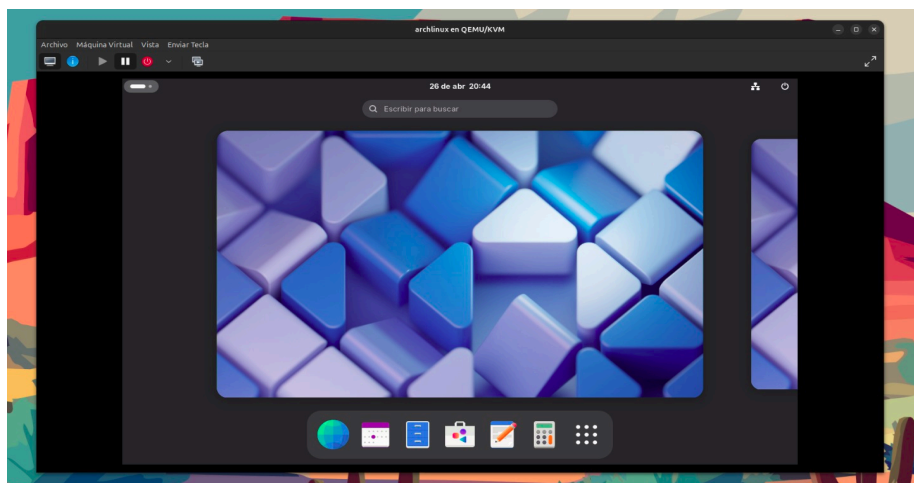
- Instalacion de virt manager

Después de instalar las VM y las Isos, descargamos las máquinas virtuales:

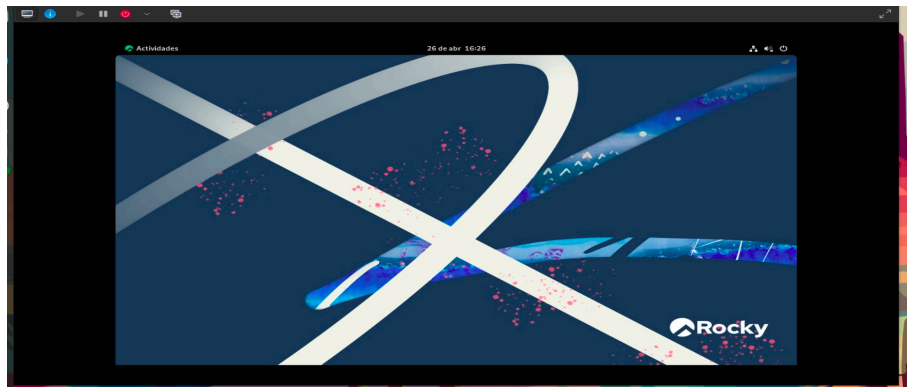
- Instalación de Manjaro:



- Instalación de Archi:

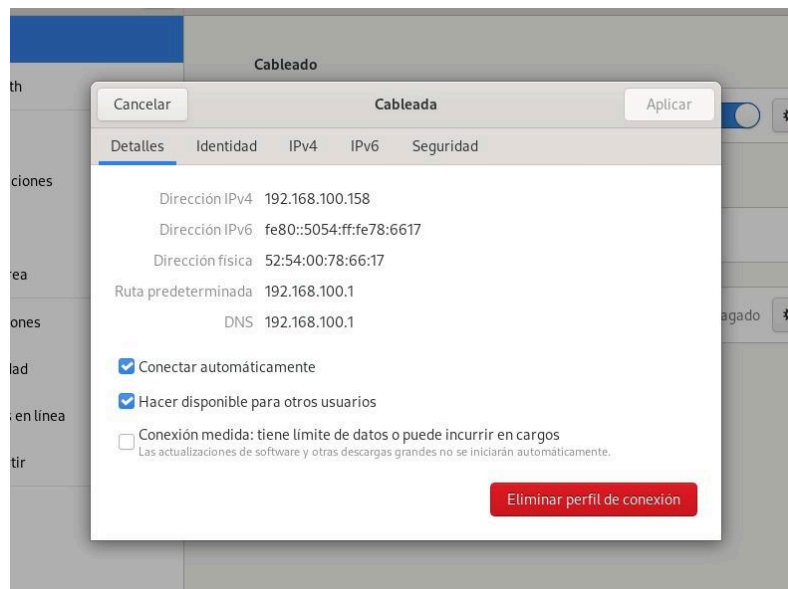


- Instalación de Rocky:

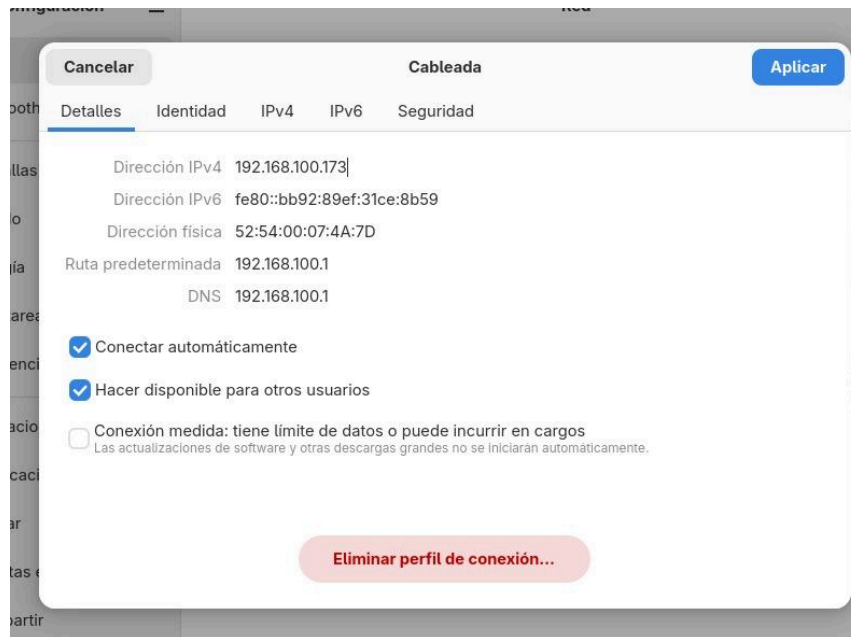


Nota: Cabe aclarar que para cada SO se tuvo que configurar el la zona horario, servidores espejos, contraseña de superusuario.

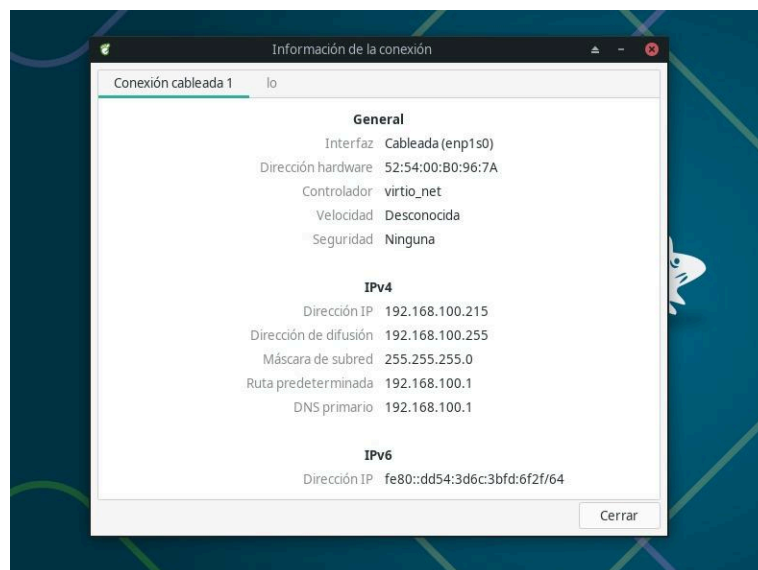
2. Creación de subred para los contenedores:



Asignación de IP para Rocky

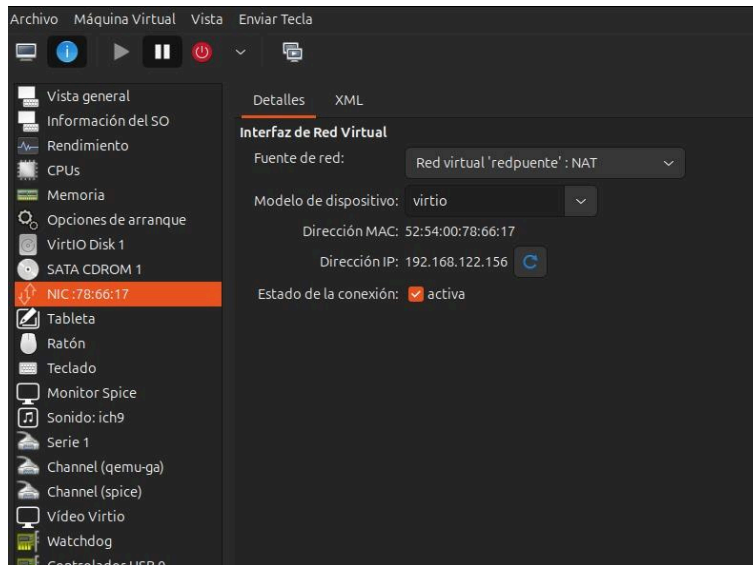


Asignación de IP para Arch Linux

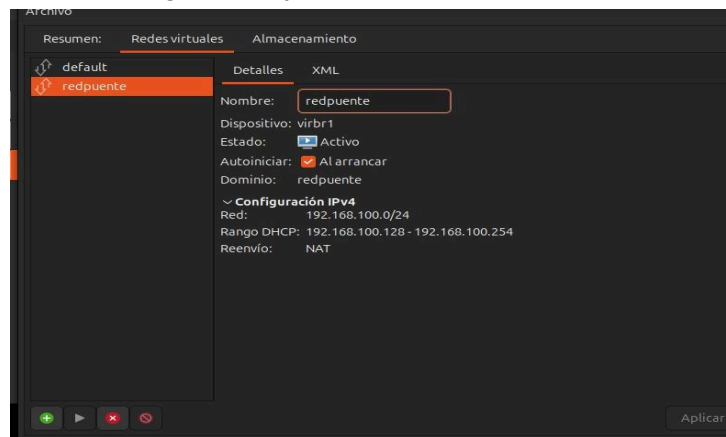


Asignación IP de Manjaro

- Asignación de Red para las **VMS**



Configuración y Creación de red para VMS



Configuración de la red de VMS

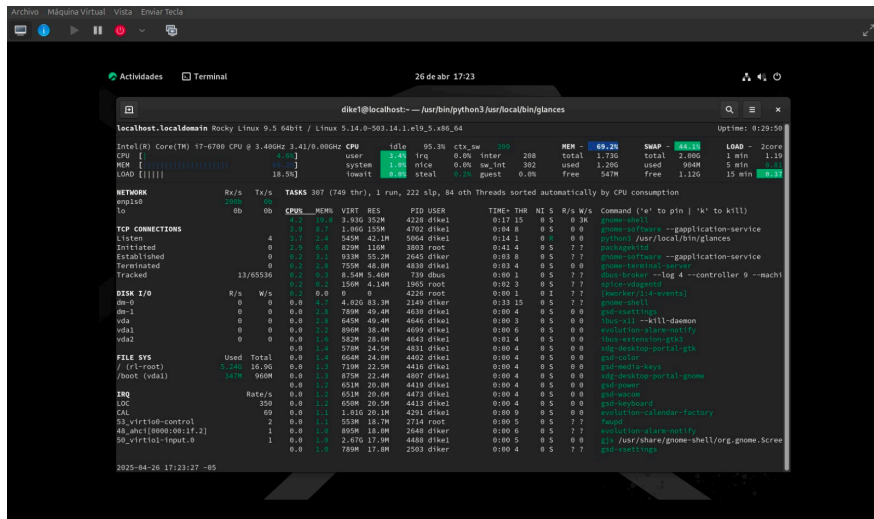
Evidencias uso de máquinas virtuales con diferentes herramientas:

```
[diker@localhost ~]$ ping -c 3 192.168.100.215
PING 192.168.100.215 (192.168.100.215) 56(84) bytes of data.
64 bytes from 192.168.100.215: icmp_seq=1 ttl=64 time=0.873 ms
64 bytes from 192.168.100.215: icmp_seq=2 ttl=64 time=0.292 ms
64 bytes from 192.168.100.215: icmp_seq=3 ttl=64 time=0.431 ms

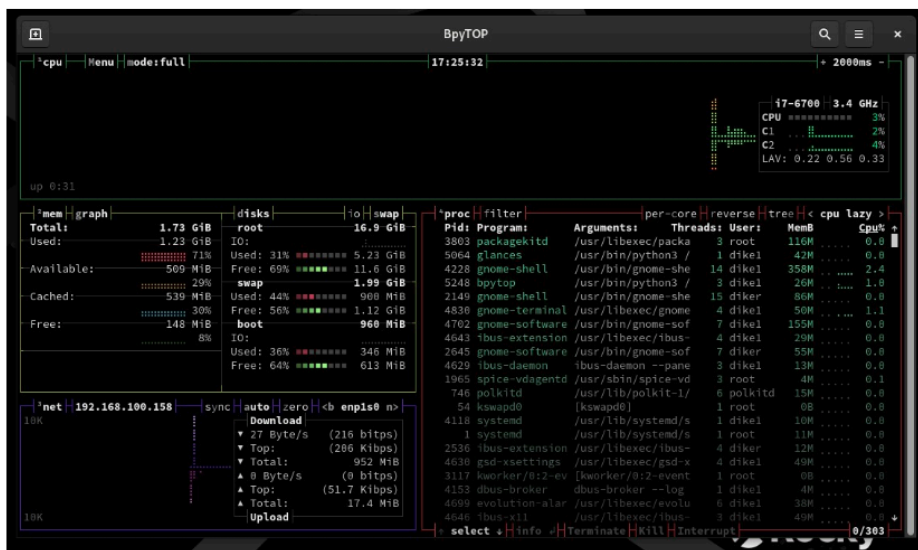
--- 192.168.100.215 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2013ms
rtt min/avg/max/mdev = 0.292/0.532/0.873/0.247 ms
[diker@localhost ~]$ ping -c 3 192.168.100.173
PING 192.168.100.173 (192.168.100.173) 56(84) bytes of data.
64 bytes from 192.168.100.173: icmp_seq=1 ttl=64 time=0.827 ms
64 bytes from 192.168.100.173: icmp_seq=2 ttl=64 time=0.452 ms
64 bytes from 192.168.100.173: icmp_seq=3 ttl=64 time=0.496 ms

--- 192.168.100.173 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2039ms
rtt min/avg/max/mdev = 0.452/0.591/0.827/0.167 ms
```

Ping de Rocky a las demás VMS



- Monitoreo de recursos con glances, en rocky
 1. La VM Rocky Linux está funcionando correctamente con entorno gráfico activo.
 2. El consumo de RAM es elevado, lo cual es esperado por el uso de GNOME y múltiples servicios de usuario.

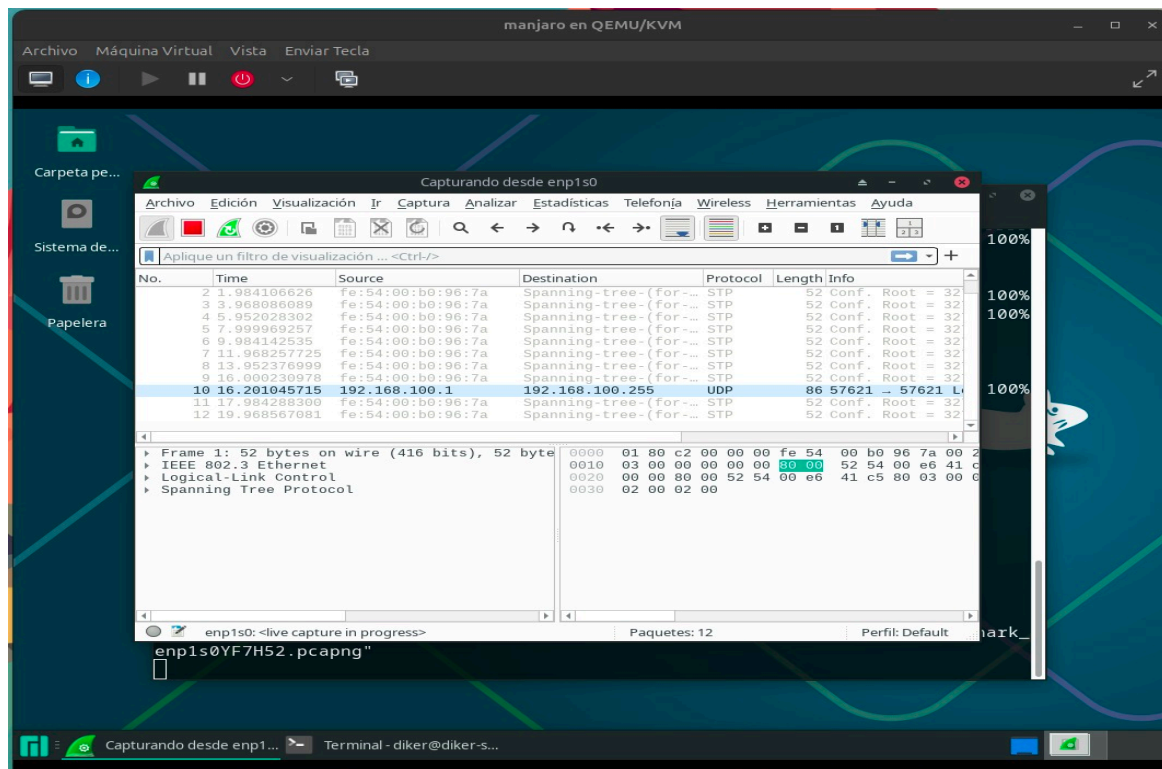


- Monitoreo de recursos con bpytop, rocky

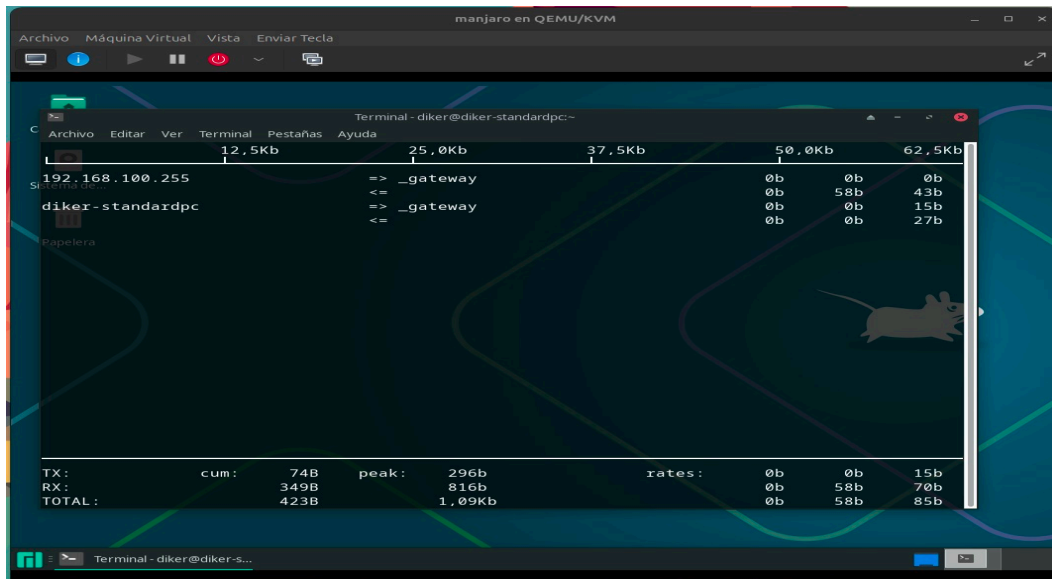
```
[dikel@localhost ~]$ sudo systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-04-26 16:44:11 -05; 42min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 889 (sshd)
      Tasks: 1 (limit: 10948)
     Memory: 332.0K
        CPU: 13ms
    CGroup: /system.slice/ssh.service
            └─889 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

abr 26 16:44:11 localhost.localdomain systemd[1]: Starting OpenSSH server daemon...
abr 26 16:44:11 localhost.localdomain sshd[889]: Server listening on 0.0.0.0 port 22.
abr 26 16:44:11 localhost.localdomain sshd[889]: Server listening on :: port 22.
abr 26 16:44:11 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
```

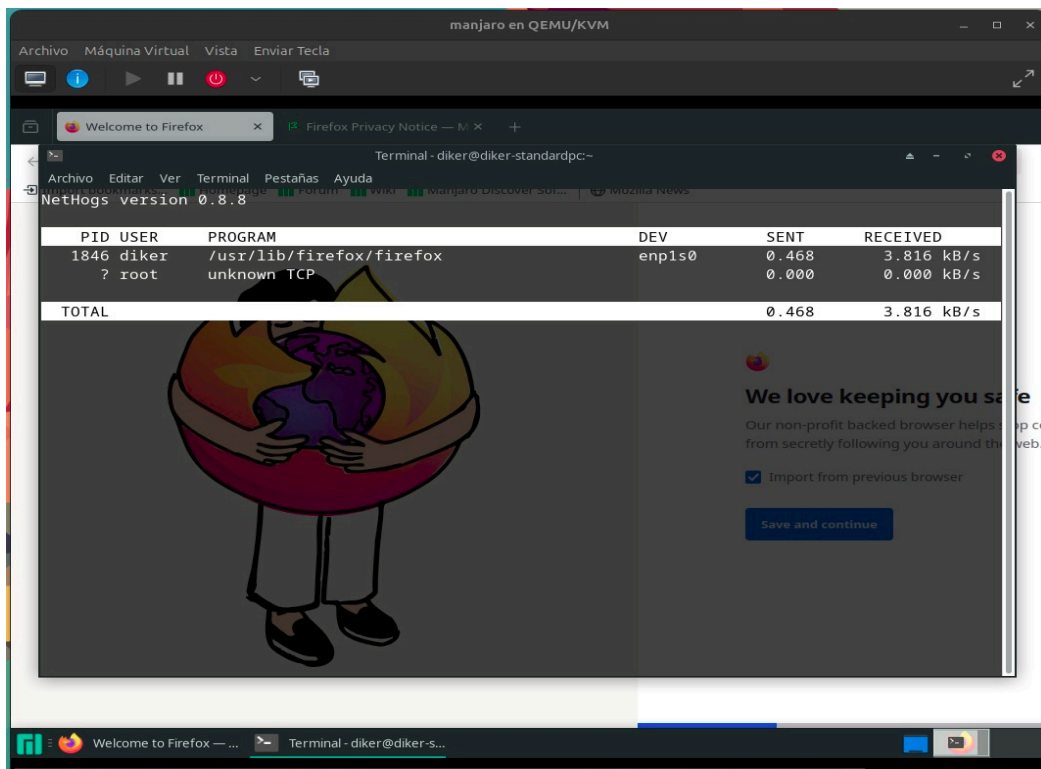
- monitoreo de un servicio sshd con systemctl



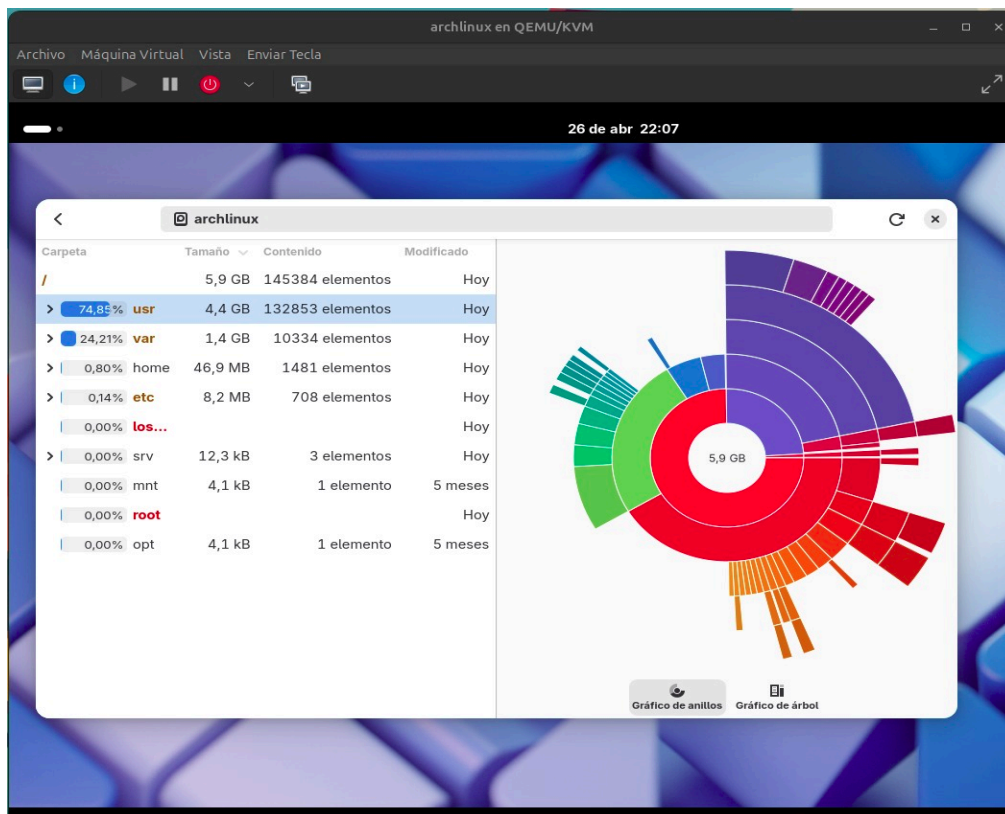
- Monitoreo de paquetes de red con wireshark, en manjaro



- iftop es un monitor en tiempo real del uso de ancho de banda por conexión. en manjaro



- nethogs agrupa el tráfico por proceso y no solo por IP. Para este se abrió Firefox, manjaro



- baobab es una aplicación gráfica muy visual que muestra el uso del disco. archi

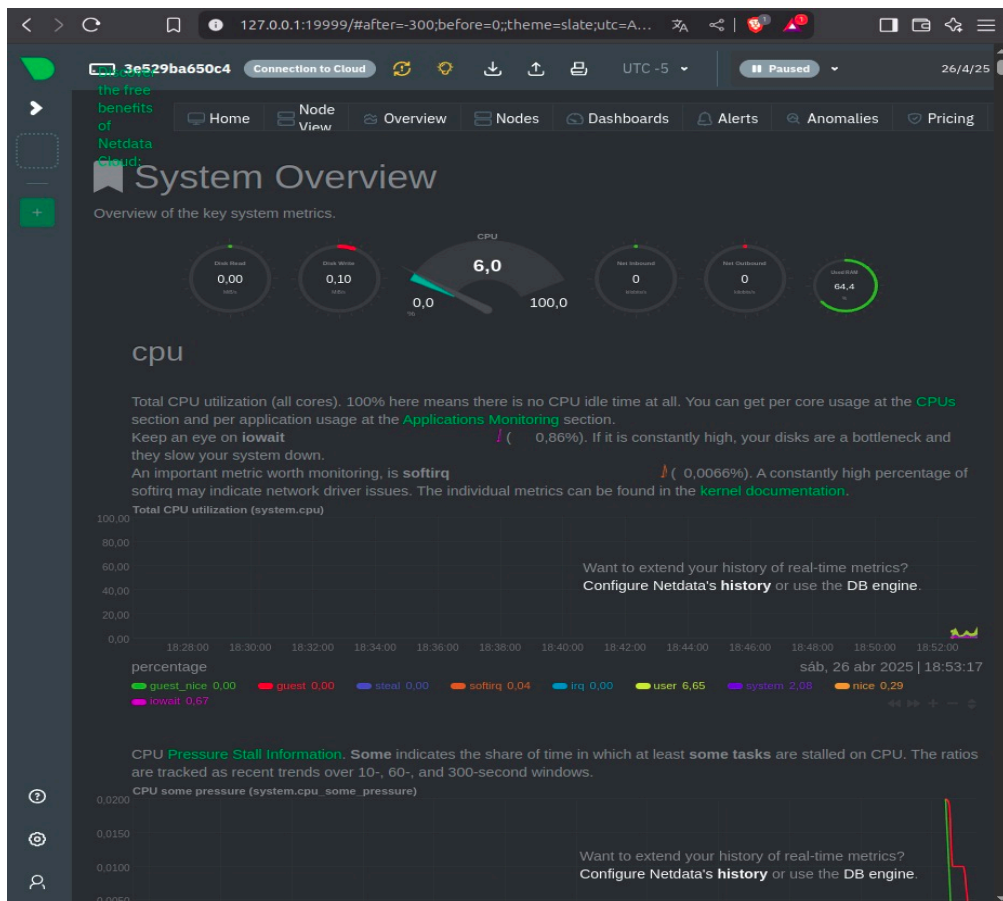
```

diker@archlinux:~/Documentos
:: ¿Continuar con la instalación? [S/n] s
:: Obteniendo los paquetes...
rsync-3.4.1-2-x86_64 383,1 KiB 852 KiB/s 00:00 [#####] 100%
(1/1) comprobando las claves del depósito [#####] 100%
(1/1) verificando la integridad de los paquetes [#####] 100%
(1/1) cargando los archivos de los paquetes [#####] 100%
(1/1) comprobando conflictos entre archivos [#####] 100%
(1/1) comprobando el espacio disponible en el ... [#####] 100%
:: Procesando los cambios de los paquetes... [#####] 100%
(1/1) instalando rsync [#####] 100%
Dependencias opcionales para rsync
python: for rsync [instalado]
:: Ejecutando los «hooks» de postinstalación...
(1/2) Reloading system manager configuration...
(2/2) Arranging ConditionNeedsUpdate...
[diker@archlinux ~]$ cd Documentos/
[diker@archlinux Documentos]$ rsync -avh /nigue/ /diker/
sending incremental file list
rsync: [sender] change_dir "/nigue" failed: No such file or directory (2)
rsync: [Receiver] mkdir "/diker" failed: Permission denied (13)
rsync error: error in file IO (code 11) at main.c(791) [Receiver-3.4.1]
[diker@archlinux Documentos]$ rsync -avh nigue diker
sending incremental file list
nigue/

sent 70 bytes received 20 bytes 188,00 bytes/sec
total size is 0 speedup is 0,00
[diker@archlinux Documentos]$

```

- Para este se crearon rutas. rsync sirve para copiar, sincronizar y hacer backups de archivos y carpetas de manera eficiente archi



- Netdata de alpine

Pasos realizados:

```

root@ker-XPS-8910: /home/ker/fedora_central
{
  "Network": "",
  "ConfigOnly": false,
  "Containers": {
    "30234c2c16a65f3dc7b3707390c6c3e710342057a5b83dd0b113f9ff93c460": {
      "Name": "deblan",
      "EndpointID": "174c850bfb275ca632373b8bc542031aaa3e70cc9f8c4b7e9c40ee3ff93c27f",
      "MacAddress": "4e:1a:68:8b:e0:b1",
      "IPv4Address": "192.168.200.3/24",
      "IPv6Address": ""
    },
    "3e529ba650c4bf17200576dfa459304965a811dcc82fb761c4e203d5e8f9628": {
      "Name": "alpine",
      "EndpointID": "4864ace7fdb0676595ca9c1ad7b91843e08e086f336c2077a814d8d7b17ed01",
      "MacAddress": "3a:3e:3f:9c:1a:c1f3",
      "IPv4Address": "192.168.200.2/24",
      "IPv6Address": ""
    },
    "b3c18749e97eeec74ab9b4f796c43be5238f1b42ce60355f032b76b01707dd6c": {
      "Name": "garuda",
      "EndpointID": "5e3d1deab9fc4d6392330962f961cf2bc984a1ebdc4d1a393a5cfd75a69f3a7",
      "MacAddress": "ea:f9:42:fc:8b:80",
      "IPv4Address": "192.168.200.4/24",
      "IPv6Address": ""
    }
  },
  "Options": {},
  "Labels": {}
}
root@ker-XPS-8910: /home/ker# docker pull fedora:latest
latest: pulling from library/fedora
0231453244d5: Pull complete
Digest: sha256:ee88ab8a5c8b78087ddccadf824767e845adc19d8cdedeb56f48521eb162b43
Status: Downloaded newer image for fedora:latest
root@ker-XPS-8910: /home/ker# docker.io/library/fedora:latest
root@ker-XPS-8910: /home/ker# mkdir fedora_central
root@ker-XPS-8910: /home/ker/fedora_central# nano Dockerfile
root@ker-XPS-8910: /home/ker/fedora_central# docker build -t fedora_central .
[+] Building 81.0s (0/6) FINISHED
=> [internal] load build definition from Dockerfile                                0.3s
=> == transferring dockerfile: 1700                                              0.0s
=> [internal] load metadata for docker.io/library/fedora:latest                  0.0s
=> [internal] load .dockerignore                                                  0.0s
=> == transferring context: 20                                                    0.0s
=> [1/2] FROM docker.io/library/fedora:latest                                    0.3s
=> == [2/2] RUN dnf update -y && dnf install -y httpd net-tools iputils python3 git && dnf clean all 75.0s
=> == exporting layers                                                            2.1s
=> == writing image sha256:34dc76a8eef3b39dea3a052a5aae1dd7b94081575bfcfcab1d20545ad1c585aa 0.0s
=> == naming to docker.io/library/fedora_central                                0.1s
root@ker-XPS-8910: /home/ker/fedora_central#

```

- Parte de fedora central
 1. Se ejecuta: docker pull fedora:latest

2. Parte inferior: Construcción de la imagen personalizada. Se edita un archivo DockerFile con: FROM fedora:latest y RUN dnf update -y && dnf install -y htop net-tools iputils iputils python3 git vim \ && dnf clean
3. Se ha creado correctamente la imagen **fedora_central**, personalizada con herramientas como htop, ping, git

```
@acfff66ae33c/

0[ ] 1[ ] 2[ ] 3[ ] 4[ ] 5[ ] 6[ ] 7[ ]
Mem: [|||||] 10.5G/15.5G Tasks: 3, 0 thr : 3 running
Swap: [|||||] Load average: 0.39 0.92 1.06
Uptime: 04:33:11

Main 370
PID USER PRI NI VIRT RES SHR S CPU%MEM% TIME+ Command
1 root 20 0 2320 1048 1440 S 0.0 0.0 0:00.02 tail -f /dev/null
7 root 20 0 4788 1824 3312 S 0.0 0.0 0:00.02 bash
27 root 20 0 6192 4652 3756 R 0.0 0.0 0:00.00 htop
```

- htop fedora

```
[root@acfff66ae33c /]# ping google.com
PING google.com (142.251.132.110) 56(84) bytes of data.
64 bytes from bog03s04-in-f14.1e100.net (142.251.132.110): icmp_seq=1 ttl=114 time=10.0 ms
64 bytes from bog03s04-in-f14.1e100.net (142.251.132.110): icmp_seq=2 ttl=114 time=11.8 ms
64 bytes from bog03s04-in-f14.1e100.net (142.251.132.110): icmp_seq=3 ttl=114 time=9.68 ms
64 bytes from bog03s04-in-f14.1e100.net (142.251.132.110): icmp_seq=4 ttl=114 time=21.3 ms
```

- ping google.com fedora

```
"ConfigOnly": false,
"Containers": {
  "30234c2cf16ab65f3dc7b3707390c6c3e710342857a5b83dd0b113f9ff93c460": {
    "Name": "debian",
    "EndpointID": "174c856bfba275ca632373b8bc542031aaa3e76cc9f8cdb7e9c46ee3ff93c27f",
    "MacAddress": "4e:1a:68:8b:e0:b1",
    "IPv4Address": "192.168.200.3/24",
    "IPv6Address": ""
  },
  "3e529ba650c4bf172080576dfa459304965a811dcc82fb761c4e283d5e8f9628": {
    "Name": "alpine",
    "EndpointID": "4864ace7fdb0676595ca9c1ad7b91843e08e086f336c2077a814d8d7b17ed01",
    "MacAddress": "3a:3e:3f:6c:ac:f3",
    "IPv4Address": "192.168.200.2/24",
    "IPv6Address": ""
  },
  "acfff66ae33c9165267d9dcecf4f12e220c36a13603662b64029bd54e81c9ed3": {
    "Name": "fedora_central",
    "EndpointID": "612fcaa0edaf8fa04b6f90f49182853c1d8d92c74ab4ea6c6df317a44d514526",
    "MacAddress": "52:bb:f7:af:a4:13",
    "IPv4Address": "192.168.200.5/24",
    "IPv6Address": ""
  },
  "b3c18749e97eec74ab9b4f796c43be5238f1b42ce60355f032b76b01707dd6c": {
    "Name": "garuda",
    "EndpointID": "5e3d1deab9fc4d6392330962f961cf2bc984a1ebdc4d1a393a5cfd75a69f3a7",
    "MacAddress": "ea:f9:42:fc:8b:80",
    "IPv4Address": "192.168.200.4/24",
    "IPv6Address": ""
  }
},
"Options": {},
"Labels": {}
```


- ip archi 192.168.100.173
- ip manjaro 192.168.100.215
- ip rocky 192.168.100.158
- ip debian 192.168.200.3
- ip alpine 192.168.200.2
- ip garuda 192.168.200.4

```

ping -c 3 192.168.200.3 # Debian
ping -c 3 192.168.200.4 # Garuda

ping -c 3 192.168.100.158 # Rocky
ping -c 3 192.168.100.173 # Arch
ping -c 3 192.168.100.215 # Manjaro
PING 192.168.200.2 (192.168.200.2) 56(84) bytes of data.
64 bytes from 192.168.200.2: icmp_seq=1 ttl=64 time=0.120 ms
64 bytes from 192.168.200.2: icmp_seq=2 ttl=64 time=0.033 ms
64 bytes from 192.168.200.2: icmp_seq=3 ttl=64 time=0.056 ms

--- 192.168.200.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2006ms
rtt min/avg/max/ndev = 0.033/0.069/0.120/0.036 ms
PING 192.168.200.3 (192.168.200.3) 56(84) bytes of data.
64 bytes from 192.168.200.3: icmp_seq=1 ttl=64 time=0.125 ms
64 bytes from 192.168.200.3: icmp_seq=2 ttl=64 time=0.043 ms
64 bytes from 192.168.200.3: icmp_seq=3 ttl=64 time=0.060 ms

--- 192.168.200.3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2045ms
rtt min/avg/max/ndev = 0.043/0.076/0.125/0.035 ms
PING 192.168.200.4 (192.168.200.4) 56(84) bytes of data.
64 bytes from 192.168.200.4: icmp_seq=1 ttl=64 time=0.070 ms
64 bytes from 192.168.200.4: icmp_seq=2 ttl=64 time=0.054 ms
64 bytes from 192.168.200.4: icmp_seq=3 ttl=64 time=0.047 ms

--- 192.168.200.4 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2046ms
rtt min/avg/max/ndev = 0.047/0.059/0.070/0.013 ms
PING 192.168.100.158 (192.168.100.158) 56(84) bytes of data.
64 bytes from 192.168.100.158: icmp_seq=1 ttl=63 time=0.396 ms
64 bytes from 192.168.100.158: icmp_seq=2 ttl=63 time=0.403 ms
64 bytes from 192.168.100.158: icmp_seq=3 ttl=63 time=0.323 ms

--- 192.168.100.158 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2046ms
rtt min/avg/max/ndev = 0.323/0.374/0.403/0.036 ms
PING 192.168.100.173 (192.168.100.173) 56(84) bytes of data.
64 bytes from 192.168.100.173: icmp_seq=1 ttl=63 time=0.230 ms
64 bytes from 192.168.100.173: icmp_seq=2 ttl=63 time=0.320 ms
64 bytes from 192.168.100.173: icmp_seq=3 ttl=63 time=0.334 ms

--- 192.168.100.173 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2046ms
rtt min/avg/max/ndev = 0.230/0.294/0.334/0.046 ms
PING 192.168.100.215 (192.168.100.215) 56(84) bytes of data.
64 bytes from 192.168.100.215: icmp_seq=1 ttl=63 time=0.299 ms
64 bytes from 192.168.100.215: icmp_seq=2 ttl=63 time=0.269 ms
64 bytes from 192.168.100.215: icmp_seq=3 ttl=63 time=0.285 ms

--- 192.168.100.215 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2044ms
rtt min/avg/max/ndev = 0.269/0.284/0.299/0.012 ms
[root@acff66ae33c /]#

```

- ping fedora con los contenedores y MV
 1. Se están realizando pruebas de conectividad con el comando ping desde una máquina a diversas IPs dentro de las subredes 192.168.200.x y 192.168.100.x, que corresponden a contenedores Docker y máquinas virtuales (VMs) del proyecto.

IP	Sistema	Resultado	Latencia promedio
192.168.200.3	Debian (contenedor)	✓ 3/3 respuestas	~0.03 ms
192.168.200.4	Garuda (contenedor)	✓ 3/3 respuestas	~0.05 ms
192.168.200.2	Alpine (contenedor)	✓ 3/3 respuestas	~0.05 ms
192.168.100.158	Rocky (VM)	✓ 3/3 respuestas	~0.30 ms
192.168.100.173	Arch (VM)	✓ 3/3 respuestas	~0.33 ms
192.168.100.215	Manjaro (VM)	✓ 3/3 respuestas	~0.29 ms

```

[acfff66ae33c ~]# nmap -sn 192.168.200.0/24
Starting Nmap 7.92 ( https://nmap.org ) at 2025-04-27 01:34 UTC
Nmap scan report for ker-XPS-8910 (192.168.200.1)
Host is up (0.000030s latency).
MAC Address: C2:05:D4:F5:F8:AE (Unknown)
Nmap scan report for alpine.containers-net (192.168.200.2)
Host is up (0.0000090s latency).
MAC Address: 3A:3E:3F:6C:AC:F3 (Unknown)
Nmap scan report for debian.containers-net (192.168.200.3)
Host is up (0.000010s latency).
MAC Address: 4E:1A:68:8B:E0:B1 (Unknown)
Nmap scan report for garuda.containers-net (192.168.200.4)
Host is up (0.000019s latency).
MAC Address: EA:F9:42:FC:8B:80 (Unknown)
Nmap scan report for acfff66ae33c (192.168.200.5)
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.25 seconds
[acfff66ae33c ~]#

```

- escaneo de docker ip 192.168.200.
 1. El escaneo confirma que las VMs y contenedores están encendidos y accesibles dentro de las subredes 192.168.100.x y 192.168.200.x.

```

Nmap scan report for garuda.containers-net (192.168.200.4)
Host is up (0.000019s latency).
MAC Address: EA:F9:42:FC:8B:80 (Unknown)
Nmap scan report for acfff66ae33c (192.168.200.5)
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.25 seconds
[acfff66ae33c ~]# nmap -sn 192.168.100.0/24
Starting Nmap 7.92 ( https://nmap.org ) at 2025-04-27 01:35 UTC
Nmap scan report for ker-XPS-8910 (192.168.100.1)
Host is up (0.000048s latency).
Nmap scan report for 192.168.100.158
Host is up (0.00028s latency).
Nmap scan report for 192.168.100.173
Host is up (0.00029s latency).
Nmap scan report for 192.168.100.215
Host is up (0.000096s latency).
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.64 seconds
[acfff66ae33c ~]#

```

- escaneo de dispositivos en VMs ip 192.168.100
 1. Se ejecuta un escaneo de red tipo "ping scan" (sin escaneo de puertos) con el comando: `nmap -sn 192.168.100.0/24`
 2. Esto permite detectar qué dispositivos están activos en el rango 192.168.100.0/24

3. El escaneo confirma que las VMs y contenedores están encendidos y accesibles dentro de las subredes 192.168.100.x y 192.168.200.x.

```
Starting Nmap 7.92 ( https://nmap.org ) at 2025-04-27 01:30 UTC
Nmap scan report for alpine.containers-net (192.168.200.2)
Host is up (0.0000060s latency).
All 1000 scanned ports on alpine.containers-net (192.168.200.2) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 3A:3E:3F:6C:AC:F3 (Unknown)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.40 seconds
Starting Nmap 7.92 ( https://nmap.org ) at 2025-04-27 01:30 UTC
Nmap scan report for debian.containers-net (192.168.200.3)
Host is up (0.0000090s latency).
All 1000 scanned ports on debian.containers-net (192.168.200.3) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 4E:1A:6B:BB:E0:B1 (Unknown)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.39 seconds
Starting Nmap 7.92 ( https://nmap.org ) at 2025-04-27 01:30 UTC
Nmap scan report for garuda.containers-net (192.168.200.4)
Host is up (0.0000060s latency).
All 1000 scanned ports on garuda.containers-net (192.168.200.4) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: EA:F9:42:FC:8B:80 (Unknown)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds
Starting Nmap 7.92 ( https://nmap.org ) at 2025-04-27 01:30 UTC
Nmap scan report for acfff66ae33c (192.168.200.5)
Host is up (0.0000040s latency).
All 1000 scanned ports on acfff66ae33c (192.168.200.5) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.33 seconds
Starting Nmap 7.92 ( https://nmap.org ) at 2025-04-27 01:30 UTC
Nmap scan report for 192.168.100.150
Host is up (0.000205s latency).
Not shown: 999 filtered tcp ports (no-response), 8 filtered tcp ports (admin-prohibited)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.7 (protocol 2.0)
9090/tcp  closed zeus-admin

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.61 seconds
Starting Nmap 7.92 ( https://nmap.org ) at 2025-04-27 01:30 UTC
Nmap scan report for 192.168.100.173
Host is up (0.000046s latency).
All 1000 scanned ports on 192.168.100.173 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.45 seconds
Starting Nmap 7.92 ( https://nmap.org ) at 2025-04-27 01:30 UTC
Nmap scan report for 192.168.100.215
Host is up (0.000030s latency).
```

- Escaneo de puertos activos por IP.
 1. Se están realizando escaneos de puertos mediante el comando nmap a diferentes direcciones IP de contenedores y máquinas virtuales en una red local Docker y bridge (ej. 192.168.200.x, 192.168.100.x).
 2. Cada bloque representa el resultado de un escaneo a una IP distinta, correspondiente a Alpine, Debian, Garuda, Rocky, Arch, Manjaro y Fedora.

```

root 55 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 57 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 59 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 61 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 63 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 65 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 67 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 69 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 71 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 81 0.0 0.0 0 0 7 Z Apr26 0:00 [gpgconf] <defunct>
root 84 0.0 0.0 0 0 7 Z Apr26 0:00 [gpgconf] <defunct>
root 86 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 88 0.0 0.0 0 0 7 Z Apr26 0:00 [gpgsm] <defunct>
root 90 0.0 0.0 0 0 7 Z Apr26 0:00 [gpgconf] <defunct>
root 92 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 94 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 96 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 98 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 100 0.0 0.0 0 0 7 Z Apr26 0:00 [gpg] <defunct>
root 179 0.0 0.0 5244 4200 pts/0 Ss 01:41 0:00 bash
root 195 0.0 0.0 0 0 pts/0 Z 01:42 0:00 [gpgconf] <defunct>
root 197 0.0 0.0 0 0 pts/0 Z 01:42 0:00 [gpgconf] <defunct>
root 199 0.0 0.0 0 0 pts/0 Z 01:42 0:00 [gpg] <defunct>
root 201 0.0 0.0 0 0 pts/0 Z 01:42 0:00 [gpgsm] <defunct>
root 203 0.0 0.0 0 0 pts/0 Z 01:42 0:00 [gpgconf] <defunct>
root 205 0.0 0.0 0 0 pts/0 Z 01:42 0:00 [gpg] <defunct>
root 207 0.0 0.0 0 0 pts/0 Z 01:42 0:00 [gpg] <defunct>
root 209 0.0 0.0 0 0 pts/0 Z 01:42 0:00 [gpg] <defunct>
root 213 0.0 0.0 7120 3900 pts/0 R+ 01:43 0:00 ps aux
[root@bc18749e97e /]# exit
exit
root@ker-XPS-0918:/home/ker/fedora_central# docker exec -it alpine bash
3e529ba650c4:/# apk update && apk add procps
fetch https://dl-cdn.alpinelinux.org/alpine/v3.21/main/x86_64/APKINDEX.tar.gz
fetch https://dl-cdn.alpinelinux.org/alpine/v3.21/community/x86_64/APKINDEX.tar.gz
v3.21.3-407-g0084c0899a2 [https://dl-cdn.alpinelinux.org/alpine/v3.21/main]
v3.21.3-411-g79994caca7e [https://dl-cdn.alpinelinux.org/alpine/v3.21/community]
OK: 25397 distinct packages available
(1/3) Installing libintl (0.22.5-r0)
(2/3) Installing libproc2 (4.0.4-r2)
(3/3) Installing procps-ng (4.0.4-r2)
Executing busybox-1.37.0-r12.trigger
OK: 67 MiB in 92 packages
3e529ba650c4:/# ps aux
USER      PID  CPU  MEM      VSZ    RSS  TTY      STAT  START   TIME  COMMAND
root         1  0.0  0.0  1628    772 ?        Ss   Apr26   0:00  tail -f /dev/null
netdata    21  0.0  0.3 530608 53540 ?        Ssl  Apr26   0:30  netdata -D
netdata    26  0.0  0.0  0      0 ?        ZN   Apr26   0:00  [spawn-plugins] <defunct>
netdata   282  0.0  0.0  0      0 ?        ZN   Apr26   0:00  [spawn-setns] <defunct>
netdata   639  0.0  0.0  0      0 ?        ZN   Apr26   0:00  [sh] <defunct>
root     643  0.1  0.0  2624  2092 pts/0  Ss   01:43   0:00  bash
root     653  0.0  0.0  2524  1072 pts/0  R+   01:43   0:00  ps aux
3e529ba650c4:/# exit
exit
root@ker-XPS-0918:/home/ker/fedora_central#

```

```

bash: apt: command not found
[root@bc18749e97e /]# df -h
bash: df: command not found
[root@bc18749e97e /]# apk update && apk add procps
bash: apk: command not found
[root@bc18749e97e /]# pacman -Sy --noconfirm procps-ng
:: Synchronizing package databases...
core downloading...
extra downloading...
warning: procps-ng-4.0.5-3 is up to date -- reinstalling
resolving dependencies...
looking for conflicting packages...

Package (1)      Old Version  New Version  Net Change  Download Size
core/procps-ng  4.0.5-3      4.0.5-3      0.00 MiB    0.05 MiB

Total Download Size: 0.05 MiB
Total Installed Size: 2.36 MiB
Net Upgrade Size: 0.00 MiB

:: Proceed with installation? [Y/n]
:: Retrieving packages...
procps-ng-4.0.5-3-x86_64 downloading...
checking keyring...
checking package integrity...
loading package files...
checking for file conflicts...
:: Processing package changes...
reinstalling procps-ng...
:: Running post-transaction hooks...
(1/1) Arming ConditionNeedsUpdate...
[root@bc18749e97e /]# ps aux
USER      PID  CPU  MEM      VSZ    RSS  TTY      STAT  START   TIME  COMMAND
root         1  0.0  0.0  3132   2160 ?        Ss   Apr26   0:00  tail -f /dev/null
root        21  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpgconf] <defunct>
root        23  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpgconf] <defunct>
root        25  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpg] <defunct>
root        27  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpgsm] <defunct>
root        29  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpgconf] <defunct>
root        31  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpg] <defunct>
root        33  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpg] <defunct>
root        35  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpg] <defunct>
root        45  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpgconf] <defunct>
root        47  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpgconf] <defunct>
root        49  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpg] <defunct>
root        51  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpgsm] <defunct>
root        53  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpgconf] <defunct>
root        55  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpg] <defunct>
root        57  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpg] <defunct>
root        59  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpg] <defunct>
root        61  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpg] <defunct>
root        63  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpg] <defunct>
root        65  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpg] <defunct>
root        67  0.0  0.0  0      0 ?        Z    Apr26   0:00  [gpg] <defunct>

```



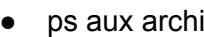
```

root@ker-XPS-8910:/home/ker/fedora_central# docker exec -it fedora_central bash
[root@acff66ae33c /]# ps aux
bash: ps: command not found
[root@acff66ae33c /]# dnf install -y procps
Updating and loading repositories:
Repositories loaded.
Package Arch Version Repository
Installing:
procps-ng x86_64 4.0.4-6.fc42 fedora 1.0
Transaction Summary:
Installing: 1 package
Total size of inbound packages is 365 KiB. Need to download 365 KiB.
After this operation, 1 MiB extra will be used (install 1 MiB, remove 0 B).
[1/1] procps-ng-0:4.0.4-6.fc42.x86_64 100% | 1.8 MiB/s | 365.3 KiB | 00m
[1/1] Total 100% | 379.4 KiB/s | 365.3 KiB | 00m
Running transaction
[1/3] Verify package files 100% | 333.0 B/s | 1.0 B | 00m
[2/3] Prepare transaction 100% | 26.0 B/s | 1.0 B | 00m
[3/3] Installing procps-ng-0:4.0.4-6.fc42.x86_64 100% | 830.5 KiB/s | 1.0 MiB | 00m
Complete!
[root@acff66ae33c /]# ps aux
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND
root 1 0.0 0.0 2596 1448 ? Ss 00:21 0:00 tail -f /dev/null
root 27 0.0 0.0 0 0 ? Z 00:22 0:00 [htop] <defunct>
root 28 0.0 0.0 0 0 ? Z 00:22 0:00 [ping] <defunct>
root 110 0.0 0.0 4788 3896 pts/0 Ss 01:38 0:00 bash
root 136 0.0 0.0 6892 3996 pts/0 R+ 01:40 0:00 ps aux
[root@acff66ae33c /]# exit
exit
root@ker-XPS-8910:/home/ker/fedora_central# docker exec -it debian bash
root@30234c2cf16a:/# apt update && apt install -y procps
Hit:1 http://deb.debian.org/debian stable InRelease
Hit:2 http://deb.debian.org/debian stable-updates InRelease
Hit:3 http://deb.debian.org/debian-security stable-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
1 package can be upgraded. Run 'apt list --upgradable' to see it.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
libproc2-0 psmisc
The following NEW packages will be installed:
libproc2-0 procps psmisc
0 upgraded, 3 newly installed, 0 to remove and 1 not upgraded.
Need to get 1030 kB of archives.
After this operation, 3310 kB of additional disk space will be used.
Get:1 http://deb.debian.org/debian stable/main amd64 libproc2-0 amd64 2:4.0.2-3 [62.8 kB]
Get:2 http://deb.debian.org/debian stable/main amd64 procps amd64 2:4.0.2-3 [709 kB]
Get:3 http://deb.debian.org/debian stable/main amd64 psmisc amd64 23.6-1 [259 kB]
Fetched 1030 kB in 0s (4707 kB/s)

```

Descripción de lo que sucede en la imágenes:

- Acceso al contenedor Alpine
 - docker exec -it alpine bash
- Intento fallido con dnf (o con apt)
- Comando correcto en Alpine: instalar procps-ng
 - apk update # actualiza el índice de paquetes
 - apk add procps-ng
- Verificación en Alpine
- Desde el shell se usa Pacman (Arch) para instalar lo mismo:
 Instalación en otro contenedor (Arch/Garuda)
 - pacman -S --noconfirm procps-ng
 - ps aux confirma la instalación mostrando la lista de procesos
- Resultado final: **Todos los contenedores** (Fedora, Debian, Alpine, Garuda/Arch) ya disponen de las utilidades de monitoreo de procesos (ps, top, htop).



- 1.En el terminal (ubicado en el directorio ~/Documentos) se ejecutó el comando: ps aux
- 2.ps aux muestra **todos los procesos** en ejecución, con información clave:

Columna	Significado (ejemplos en la captura)
USER	Usuario propietario del proceso (<code>diker</code> , <code>root</code>)
PID	ID del proceso (1367, 1469, 1591...)
%CPU / %MEM	Uso de CPU y memoria de cada proceso (casi 0 % en la mayoría)
VSZ / RSS	Memoria virtual y residente en KB
STAT	Estado (S: sleep, R: running, Z: zombie, etc.)
START	Hora de inicio (abr 26 / 01:49)
TIME	Tiempo de CPU consumido
COMMAND	Ejecutable (p. ej. <code>/usr/lib/xdg-desktop-portal</code> , <code>/usr/lib/pulseaudio</code> , <code>ps</code> , <code>bash</code>)

3. Se muestra todo el listado de procesos y confirma que el estado operativo del sistema con baja carga de recursos

```

manjaro en QEMU/KVM
Archivo  Máquina Virtual  Vista  Limpiar Tecla
Carpeta pe...
Sistema de...
Papelera

terminal - diker@tiker-standardpc-
Archivo  Editar  Ver  Terminal  Pestañas  Ayuda
diker    923    0.0  0.1 157956 7156 ?    Sls  19:07  0:00 /usr/bin/gpg-
diker    925    0.0  5.5 1074000 221424 ?    Sl   19:07  0:02 x11wm4
diker    940    0.0  0.1 163784 5596 ?    Ssl  19:07  0:00 /usr/lib/dcon
diker    941    0.0  1.4 553820 60064 ?    Sl   19:07  0:00 xfce4-panel
diker    949    0.0  0.3 103648 15612 ?    S<sl 19:07  0:00 /usr/bin/pipe
diker    950    0.0  0.9 763800 37884 ?    S<sl 19:07  0:00 /usr/bin/wire
diker    951    0.0  0.3 101168 12476 ?    S<sl 19:07  0:00 /usr/bin/pipe
diker    953    0.0  0.9 457512 39604 ?    Sl   19:07  0:00 Thunar --daem
rtkit    956    0.0  0.0 21712 3392 ?    SWSl 19:07  0:00 /usr/lib/rtki
diker    967    0.0  1.5 478468 61012 ?    Sl   19:07  0:02 /usr/lib/xfce
diker    979    0.0  0.9 383656 39996 ?    Sl   19:07  0:00 /usr/lib/xfce
diker    980    0.0  1.3 409280 53556 ?    Sl   19:07  0:00 /usr/lib/xfce
diker    985    0.0  1.2 474360 51916 ?    Sl   19:07  0:00 /usr/lib/xfce
diker    995    0.0  2.1 668300 87404 ?    Sl   19:07  0:01 xfdesktop
diker   1003    0.0  1.3 547996 53200 ?    Ssl  19:07  0:00 /usr/lib/xfce
diker   1017    0.0  0.0 12508 2072 ?    Ssl  19:07  0:00 xcapp -e Supe
diker   1019    0.0  1.8 355528 73096 ?    Sl   19:07  0:02 xfce4-clipman
diker   1020    0.0  1.7 567620 69128 ?    Sl   19:07  0:00 /usr/bin/pyth
diker   1027    0.0  1.3 423240 55236 ?    Sl   19:07  0:01 mnm_notifier
diker   1028    0.0  1.6 648288 66812 ?    Sl   19:07  0:00 nm-applet
diker   1029    0.0  1.5 353124 60780 ?    Sl   19:07  0:00 pamac_tray
diker   1032    0.0  0.9 242268 37924 ?    Sl   19:07  0:00 /usr/lib/polk
diker   1045    0.0  0.9 383416 39984 ?    Sl   19:07  0:00 xfce4-power-m
diker   1050    0.0  1.4 260344 56512 ?    Sl   19:07  0:00 xfce4-screens

```

- `ps aux manjaro`

1. Elementos que se observan con el comando “ps aux” en manjaro:

Columna	Lo que muestra en la captura
USER	Casi todos los procesos pertenecen al usuario normal <code>diker</code> (sesión gráfica).
PID	Identificadores de procesos (923, 941, 948, ... 1045).
%CPU / %MEM	La mayor parte tiene valores muy bajos ($\approx 0\%$) $\rightarrow$ la VM está en reposo.
STAT	Muchos procesos están en modo S (sleep), algunos Sl (sleep con memoria mapeada), indicando poca actividad.
COMMAND	Procesos característicos de un escritorio XFCE: <code>/usr/bin/gpg</code> <code>/usr/lib/xfce4/...</code> (panel, compositor, xfsettingsd, xfsession) <code>/usr/lib/at-spi2-...</code> (accesibilidad) <code>panel-14-whiskermenu</code> (lanzador de aplicaciones) <code>xfce4-screensaver</code> (protector de pantalla).

```

root@ker-XPS-8910:/home/ker/fedora_central# docker exec -it fedora_central bash
[root@acfff66ae33c /]# netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.11:34081        0.0.0.0:*               LISTEN      -
udp        0      0 127.0.0.11:33306       0.0.0.0:*               -
[root@acfff66ae33c /]# exit
exit
root@ker-XPS-8910:/home/ker/fedora_central# docker exec -it debian bash
root@30234c2cf16a:/# netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.11:41647       0.0.0.0:*               LISTEN      -
udp        0      0 127.0.0.11:41125       0.0.0.0:*               -
root@30234c2cf16a:/# exit
exit
root@ker-XPS-8910:/home/ker/fedora_central# docker exec -it alpine bash
3e529ba650c4:/# netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:19999           0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:8125          0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.11:43693       0.0.0.0:*               LISTEN      -
tcp6       0      0 :::8125                 :::*                     LISTEN      -
tcp6       0      0 :::19999                :::*                     LISTEN      -
udp        0      0 127.0.0.11:51488       0.0.0.0:*               -
udp        0      0 127.0.0.1:8125         0.0.0.0:*               -
udp6       0      0 :::8125                 :::*                     -
3e529ba650c4:/# exit
exit
root@ker-XPS-8910:/home/ker/fedora_central# docker exec -it garuda bash
[root@b3c18749e97e /]# netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.11:41095       0.0.0.0:*               LISTEN      -
udp        0      0 127.0.0.11:43342       0.0.0.0:*               -
[root@b3c18749e97e /]#

```

- **docker netstat -tuln**
 1. En esta captura se ejecuta el comando `netstat -tuln` dentro de cada contenedor Docker para mostrar puertos en escucha y servicios activos por contenedor.
 2. Todos los contenedores tienen conectividad básica y servicios en escucha, lo cual valida su integración en la red Docker.

```

0[ ] 1[ ] 2[ ] 3[ ] 4[ ] 5[ ] 6[ ] 7[ ]
Mem[ ] 11.1G/15.5G Tasks: 5, 0 thr ; 1 running
Swp[ ] Load average: 0.32 0.44 0.54
Uptime: 06:09:46

Main I/O
PID USER PRI NI VIRT RES SHR S CPU%MEM% TIME+ Command
1 root 20 0 2596 1448 1448 S 0.0 0.0 0:00.20 tail -f /dev/null
27 root 20 0 0 0 0 Z 0.0 0.0 0:00.01 htop
28 root 20 0 0 0 0 Z 0.0 0.0 0:00.01 ping
195 root 20 0 4788 3856 3344 S 0.0 0.0 0:00.02 bash
214 root 20 0 6192 4872 3976 R 0.0 0.0 0:00.01 htop

```

- htop debian
 1. Terminal ejecutando htop dentro del contenedor fedora_central.
 2. Tiempo de actividad (uptime): 6h 9m.
 3. Uso de recursos: RAM Y CPU

```

0[ ] 1[ ] 2[ ] 3[ ] 4[ ] 5[ ] 6[ ] 7[ ]
Mem[ ] 11.1G/15.5G Tasks: 7, 82 thr ; 0 running
Swp[ ] Load average: 0.72 0.53 0.56
Uptime: 06:10:50

Main I/O
PID USER PRI NI VIRT RES SHR S CPU%MEM% TIME+ Command
689 root 20 0 2212 1952 1056 R 0.0 0.0 0:00.04 htop
1 root 20 0 1628 772 772 S 0.0 0.0 0:00.29 tail -f /dev/null
25 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.06 netdata -D
26 netdata 39 19 0 0 0 Z 0.0 0.0 0:00.02 spawn-plugins
27 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.00 netdata -D
185 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.78 netdata -D
186 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
187 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.00 netdata -D
188 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.00 netdata -D
189 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.02 netdata -D
190 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.00 netdata -D
191 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
192 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.00 netdata -D
193 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
194 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
195 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
196 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.00 netdata -D
197 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
198 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
199 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
200 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
201 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
202 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
203 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
204 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
205 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.04 netdata -D
206 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
207 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
208 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
209 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
210 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
211 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
212 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
213 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.00 netdata -D
214 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.00 netdata -D
215 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.00 netdata -D
216 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
217 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
218 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
219 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.00 netdata -D
220 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.00 netdata -D
221 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D
222 netdata 39 19 527M 54660 15856 S 0.0 0.3 0:00.01 netdata -D

```

- htop alpine

1. Se demuestra el uso real de htop y la carga generada por Netdata.

```

0[|||||] 0.0% 0.0% 4[|||||] 0.0%
1[|||||] 0.0% 0.0% 5[|||||] 0.0%
2[|||||] 0.0% 0.0% 6[|||||] 0.0%
3[|||||] 0.0% 0.0% 7[|||||] 0.0%
Mem[|||||] 11.1G/15.5G Tasks: 67, 0 thr; 1 running
Swp[|||||] Load average: 0.40 0.47 0.54
Uptime: 06:11:26

Main I/O
PID USER PRI NI VIRT RES SHR S CPU%MEM% TIME+ Command
309 root 20 0 5596 4144 3120 R 0.7 0.0 0:00.01 htop
1 root 20 0 3132 2160 1904 R 0.0 0.0 0:00.34 tail -f /dev/null
21 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgconf
23 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgconf
25 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
27 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgsm
29 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgconf
31 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
33 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
35 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
45 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgconf
47 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgconf
49 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
51 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgsm
53 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgconf
55 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
57 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
59 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
61 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
63 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
65 root 20 0 0 0 0 Z 0.0 0.0 0:00.04 gpg
67 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
69 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
71 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
82 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgconf
84 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgconf
86 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
88 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgsm
90 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgconf
92 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
94 root 20 0 0 0 0 Z 0.0 0.0 0:00.01 gpg
96 root 20 0 0 0 0 Z 0.0 0.0 0:00.01 gpg
98 root 20 0 0 0 0 Z 0.0 0.0 0:00.01 gpg
100 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
195 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgconf
197 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgconf
199 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
201 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgsm
203 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgconf
205 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
207 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
209 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpg
227 root 20 0 0 0 0 Z 0.0 0.0 0:00.00 gpgconf

```

- htop Garuda

1. El contenedor Garuda está corriendo correctamente, aunque lleno de procesos gpgconf en estado **zombi**, sin impacto funcional, pero notable.

```

[root@acfff60ae33c ~]# tcpdump -i any -c 10
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
(listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
02:03:03.065443 eth0 0 IP ker-XPS-8910:57621 > 192.168.200.255:57621: UDP, length 44
02:03:03.917249 lo In IP localhost.52305 > localhost.33300: UDP, length 57
02:03:03.919978 lo In IP localhost.domain > localhost.52305: 54534 NXDomain 0/0/1 (57)
02:03:04.020762 lo In IP localhost.36966 > localhost.33300: UDP, length 52
02:03:04.021194 lo In IP localhost.domain > localhost.36966: 15457*5 1/0/1 PTR localhost. (75)
^Z
[2]- Stopped tcpdump -i any -c 10
[root@acfff60ae33c ~]# exit
exit
There are stopped jobs.
[root@acfff60ae33c ~]# exit
exit

5 packets captured5 packets captured

21 packets received by filter
12 packets received by filter0 packets dropped by kernel

0 packets dropped by kernel
root@ker-XPS-8910:/home/ker/fedora_central# docker exec -it debian bash
root@30234c2cf16a:/# tcpdump -i any -c 10
tcpdump: data link type LINUX_SLL2
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
(listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
02:03:33.075595 eth0 0 IP ker-XPS-8910:57621 > 192.168.200.255:57621: UDP, length 44
02:03:33.908958 lo In IP localhost.34504 > localhost.41125: UDP, length 57
02:03:33.910936 lo In IP localhost.53 > localhost.34504: 29488 NXDomain 0/0/1 (57)
02:03:33.911084 lo In IP localhost.37675 > localhost.41125: UDP, length 55
02:03:33.925461 lo In IP localhost.53 > localhost.37675: 2040*5 2/0/1 PTR ker-XPS-8910., PTR ker-XPS-8910.loc
al. (113)
02:03:34.012040 lo In IP localhost.49415 > localhost.41125: UDP, length 52
02:03:34.013325 lo In IP localhost.53 > localhost.49415: 25543*5 1/0/1 PTR localhost. (75)
^Z
[1]- Stopped tcpdump -i any -c 10
root@30234c2cf16a:/# exit
exit
There are stopped jobs.
root@30234c2cf16a:/# exit
exit

7 packets captured
13 packets received by filter
0 packets dropped by kernel
root@ker-XPS-8910:/home/ker/fedora_central#

```

- tcpdump fedora y debian
 1. Contenedor central Fedora: se ejecuta tcpdump y se muestran que se capturaron 10 paquetes, incluyendo tráfico DNS inverso
 2. Para debian, se accede con :docker exec -it debian bash
 3. Se ejecuta con: tcpdump -i any -c 10
 4. Se observa paquetes UDP y DNS inverso que confirma que está conectado correctamente a la red

```

root@ker-XPS-8910:/home/ker/fedora_central# docker exec -it alpine bash
3e529ba650c4:/# tcpdump -i any -c 10
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
02:04:40.107304 eth0 In IP ker-XPS-8910.local.58836 > 3e529ba650c4.19999: Flags [P.], seq 721117784:721118396, a
ck 29676615, win 408, options [nop,nop,TS val 3072181349 ecr 835332936], length 612
02:04:40.107606 eth0 Out IP 3e529ba650c4.19999 > ker-XPS-8910.local.58836: Flags [P.], seq 1:598, ack 612, win 24
48, options [nop,nop,TS val 835343937 ecr 3072181349], length 597
02:04:40.107620 eth0 In IP ker-XPS-8910.local.58836 > 3e529ba650c4.19999: Flags [.], ack 598, win 404, options [
nop,nop,TS val 3072181350 ecr 835343937], length 0
02:04:40.108092 lo In IP localhost.26393 > localhost.51488: UDP, length 44
02:04:40.102818 lo In IP localhost.53 > localhost.36393: 3738* 2/0/0 PTR ker-XPS-8910., PTR ker-XPS-8910.local
. (102)
02:04:40.204865 lo In IP localhost.47227 > localhost.51488: UDP, length 41
02:04:40.205461 lo In IP localhost.53 > localhost.47227: 39586* 1/0/0 PTR localhost. (64)
^Z
[!]+ Stopped tcpdump -i any -c 10
3e529ba650c4:/# exit
exit
There are stopped jobs.
3e529ba650c4:/# exit
exit

7 packets captured
11 packets received by filter
0 packets dropped by kernel
root@ker-XPS-8910:/home/ker/fedora_central# docker exec -it garuda bash
[root@b3c18749e97e /]# tcpdump -i any -c 10
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
02:05:10.078740 eth0 M IP _gateway.47546 > 239.255.255.250.ssdp: UDP, length 168
02:05:10.213303 lo In IP localhost.58264 > localhost.43342: UDP, length 57
02:05:10.226508 lo In IP localhost.domain > localhost.58264: 6263 NXDomain 0/1/1 (114)
02:05:10.276972 lo In IP localhost.59900 > localhost.43342: UDP, length 52
02:05:10.277676 lo In IP localhost.domain > localhost.59900: 58955*5 1/0/1 PTR localhost. (75)
02:05:11.079749 eth0 M IP _gateway.47546 > 239.255.255.250.ssdp: UDP, length 168
02:05:12.001784 eth0 M IP _gateway.47546 > 239.255.255.250.ssdp: UDP, length 168
02:05:13.083297 eth0 M IP _gateway.47546 > 239.255.255.250.ssdp: UDP, length 168
^Z
[!]+ Stopped tcpdump -i any -c 10
[root@b3c18749e97e /]# exit
exit
There are stopped jobs.
[root@b3c18749e97e /]# exit
exit

8 packets captured
12 packets received by filter
0 packets dropped by kernel
root@ker-XPS-8910:/home/ker/fedora_central#

```

- tcpdump alpine y garuda
 1. Se accede Alpine con: docker exec -it alpine bash
 2. Se intenta ejecutar: tcpdump -i any -c 10
 3. Resultado: 0 packets captured. No se capturó tráfico
 4. Ahora para GARuda, se accede con: docker exec -it garuda bash
 5. Se ejecuta: tcpdump -i any -c 10
 6. Como resultado se obtienen múltiples paquetes UDP, DNS y tráfico IP


```
root@localhost:/home/dike1
Nada por hacer.
¡Listo!
[root@localhost dike1]# dnf install -y htop
Última comprobación de caducidad de metadatos hecha hace 3:45:34, el sáb 26 abr 2025 17:24:57.
No hay coincidencias para el argumento: htop
Error: No se pudo encontrar ningún resultado: htop
[root@localhost dike1]# netstat -tulpn
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
PID/Program name
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
885/cupsd
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
889/sshd: /usr/sbin
tcp6       0      0 :::1:631                :::*                     LISTEN
885/cupsd
tcp6       0      0 :::1:22                  :::*                     LISTEN
889/sshd: /usr/sbin
udp        0      0 0.0.0.0:35012           0.0.0.0:*
740/avahi-daemon: r
udp        0      0 0.0.0.0:5353            0.0.0.0:*
740/avahi-daemon: r
udp        0      0 0.0.0.0:5353            0.0.0.0:*
751/chronyd
udp6       0      0 :::5353                  :::*
740/avahi-daemon: r
udp6       0      0 :::1:323                 :::*
751/chronyd
udp6       0      0 :::52591                 :::*
740/avahi-daemon: r
[root@localhost dike1]#
```

- netstat rocky
 1. Se evidencian listado de puertos, servicios y protocolos con el comando netstat -tulpn

```
root@localhost:/home/dike1
c5.8001, length 35
21:12:37.930193 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8001, length 35
21:12:38.042238 ARP, Request who-has ker-XPS-8910 tell localhost.localdomain, le
ngth 28
21:12:38.042386 ARP, Reply ker-XPS-8910 is-at 52:54:00:e6:41:c5 (oui Unknown), l
ength 28
21:12:38.050727 IP localhost.localdomain.41865 > ker-XPS-8910.domain: 10464+ PTR
? 1.100.168.192.in-addr.arpa. (44)
21:12:38.066486 IP ker-XPS-8910.domain > localhost.localdomain.41865: 10464+ 2/0
/0 PTR ker-XPS-8910., PTR ker-XPS-8910.local. (102)
21:12:38.066862 IP localhost.localdomain.43237 > ker-XPS-8910.domain: 20750+ PTR
? 158.100.168.192.in-addr.arpa. (46)
21:12:38.067000 IP ker-XPS-8910.domain > localhost.localdomain.43237: 20750 NXDo
main 0/1/0 (125)
21:12:38.186147 ARP, Request who-has localhost.localdomain tell ker-XPS-8910, le
ngth 28
21:12:38.186158 ARP, Reply localhost.localdomain is-at 52:54:00:78:66:17 (oui Un
known), length 28
21:12:39.914599 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8001, length 35
21:12:41.962290 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8001, length 35
21:12:43.946393 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8001, length 35
21:12:45.930270 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8001, length 35
21:12:47.914394 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8001, length 35
21:12:49.962704 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8001, length 35
```

- tcpdump rocky
 1. Se está ejecutando el comando tcpdump, capturando tráfico de red en tiempo real en esta máquina.
 2. Captura tramas STP 802.1d, ARP y DNS inversa para hacer un análisis completo de red

```

root@archlinux:/home/diker

[root@archlinux diker]# netstat -tulpn
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
PID/Program name
udp      0      0 192.168.100.173:3702    0.0.0.0:*
1663/python3
udp      0      0 239.255.255.250:3702   0.0.0.0:*
1663/python3
udp      0      0 0.0.0.0:39970          0.0.0.0:*
1663/python3
udp6     0      0 fe80::bb92:89ef:31:3702 :::*
1663/python3
udp6     0      0 ff02::c:3702           :::*
1663/python3
udp6     0      0 :::57948                :::*
1663/python3
[root@archlinux diker]#

```

- netstat -tulpn Arch Linux
 1. Muestra todas las conexiones activas en **escucha (LISTEN)** para protocolos TCP/UDP y los programas asociados.

```

root@archlinux:/home/diker

CPU[||||| 7.2%] Tasks: 96, 288 thr, 87 kthr; 1 running
Mem[||||| 713M/956M] Load average: 0.13 0.10 0.04
Swp[||||| 1478M/478M] Uptime: 02:23:58

Main I/O
PID USER PRI NI VIRT RES SHR S CPU%MEM% TIME+ Command
966 diker 20 0 3838M 177M 76276 S 3.3 18.6 0:29.97 /usr/bin/gnom
2591 root 20 0 8484 6256 4976 R 2.0 0.6 0:00.16 htop
1496 diker 20 0 1129M 189M 52328 S 1.3 19.8 0:37.32 /usr/bin/kgx
1 root 20 0 22440 7716 4640 S 0.0 0.8 0:01.68 /sbin/init
240 root 20 0 48572 3032 2520 S 0.0 0.3 0:00.33 /usr/lib/syst
276 root 20 0 15120 1400 1272 S 0.0 0.1 0:00.05 /usr/lib/syst
282 systemd-ti 20 0 89892 2280 2152 S 0.0 0.2 0:00.07 /usr/lib/syst
287 root 20 0 35356 940 812 S 0.0 0.1 0:00.11 /usr/lib/syst
288 systemd-ti 20 0 89892 2280 0 S 0.0 0.2 0:00.00 /usr/lib/syst
353 dbus 20 0 8924 1200 944 S 0.0 0.1 0:00.04 /usr/bin/dbus
354 dbus 20 0 6124 1704 696 S 0.0 0.2 0:00.22 dbus-broker -
355 root 20 0 406M 8684 7404 S 0.0 0.9 0:00.18 /usr/bin/Netw
356 root 20 0 10036 2952 2560 S 0.0 0.3 0:00.15 /usr/lib/syst
363 root 20 0 406M 8684 0 S 0.0 0.9 0:00.06 /usr/bin/Netw
364 root 20 0 406M 8684 0 S 0.0 0.9 0:00.00 /usr/bin/Netw
365 root 20 0 406M 8684 0 S 0.0 0.9 0:00.02 /usr/bin/Netw
372 root 20 0 303M 3476 2964 S 0.0 0.4 0:00.02 /usr/bin/gdm
377 root 20 0 303M 3476 0 S 0.0 0.4 0:00.00 /usr/bin/gdm
378 root 20 0 303M 3476 0 S 0.0 0.4 0:00.00 /usr/bin/gdm
379 root 20 0 303M 3476 0 S 0.0 0.4 0:00.01 /usr/bin/gdm
F1 Help F2 Setup F3 Search F4 Filter F5 Tree F6 SortBy F7 Nice F8 Nice F9 Kill F10 Quit

```

- htop archi
 1. Se muestra la interfaz del comando htop, una herramienta interactiva para visualizar procesos del sistema en tiempo real.
 2. SE observan datos como la CPU, Memoria RAM, Swap, Uptime, Load average

```
root@archlinux:/home/diker
02:18:41.897865 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8002, length 35
02:18:43.946112 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8002, length 35
02:18:45.929857 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8002, length 35
02:18:47.913998 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8002, length 35
02:18:49.897794 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8002, length 35
02:18:49.924672 IP _gateway.57621 > 192.168.100.255.57621: UDP, length 44
02:18:51.945707 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8002, length 35
02:18:53.929674 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8002, length 35
02:18:55.913807 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8002, length 35
02:18:57.897403 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8002, length 35
02:18:59.945407 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8002, length 35
02:19:01.929512 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8002, length 35
02:19:03.913414 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8002, length 35
02:19:05.897361 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:
c5.8002, length 35
```

- tcpdump archi

1. Se está ejecutando el comando tcpdump, capturando tráfico de red en tiempo real.
2. IP destino: 192.168.100.255, puerto 57621 (broadcast) y intento de descubrimiento de red para una DNS desde archi

```
Terminal - root@diker-standardpc:/home/diker
[diker-standardpc diker]# netstat -tulpn
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
PID/Program name
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
685/cupsd
tcp6       0      0 :::631                  :::*                   LISTEN
685/cupsd
udp        0      0 0.0.0.0:45119          0.0.0.0:*
606/avahi-daemon: r
udp        0      0 0.0.0.0:5353           0.0.0.0:*
606/avahi-daemon: r
udp6       0      0 :::38538                :::*
606/avahi-daemon: r
udp6       0      0 :::5353                 :::*
606/avahi-daemon: r
[diker-standardpc diker]#
```

- netstat manjaro

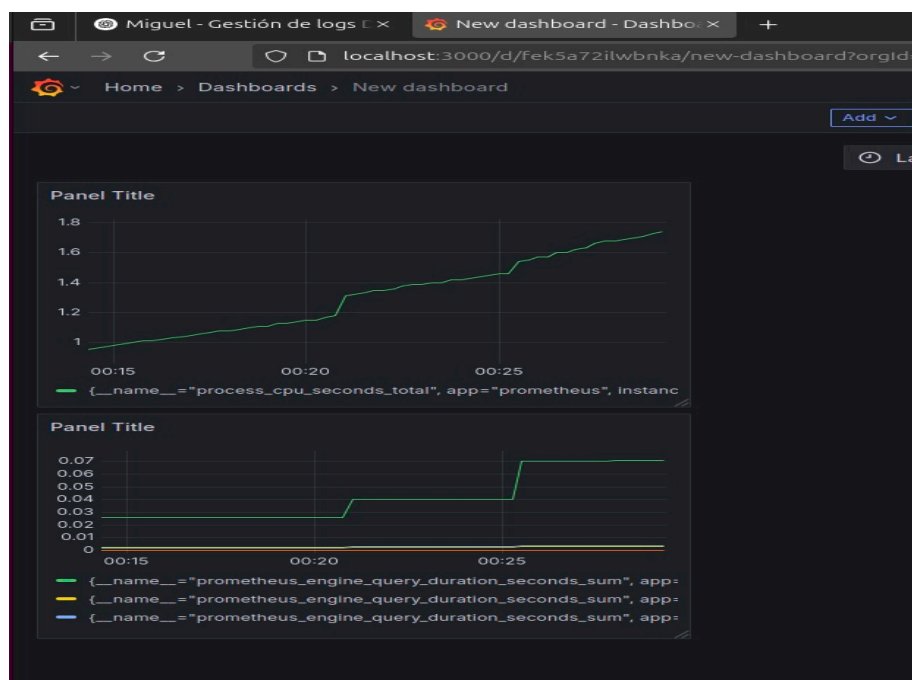
1. netstar -tulpn: Este comando muestra las conexiones activas en escucha (LISTEN), tanto TCP como UDP, junto con los puertos y programas que las manejan.

```
Terminal - root@diker-standardpc:/home/diker
h 28
21:24:12.461940 ARP, Request who-has diker-standardpc tell _gateway, length 28
21:24:12.461954 ARP, Reply diker-standardpc is-at 52:54:00:b0:96:7a (oui Unknown), length 28
21:24:13.934114 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:c5.8003, length 35
21:24:15.917967 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:c5.8003, length 35
21:24:17.966711 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:c5.8003, length 35
21:24:19.949751 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:c5.8003, length 35
21:24:21.933707 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:c5.8003, length 35
21:24:21.969833 IP _gateway.57621 > 192.168.100.255.57621: UDP, length 44
21:24:22.011040 IP diker-standardpc.46595 > _gateway.domain: 39929+ PTR? 255.100.168.192.in-addr.arpa. (46)
21:24:22.011209 IP _gateway.domain > diker-standardpc.46595: 39929 NXDomain 0/1/0 (125)
21:24:23.917765 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:c5.8003, length 35
21:24:25.901494 STP 802.1d, Config, Flags [none], bridge-id 8000.52:54:00:e6:41:c5.8003, length 35
```

- tcpdump

1. Analisis e interpretacion de lo realizado:

- La VM Manjaro realiza resoluciones ARP para comunicarse con su puerta de enlace.
- El tráfico de Spanning-Tree (STP) es típico de un adaptador puenteado en un entorno de red virtual
- Se ve una consulta DNS inversa (PTR) que la VM realiza



- monitoreo de prometheus

1.Panel 1 (arriba):

- Métrica process_cpu_seconds_total → muestra el **tiempo de CPU acumulado** por Prometheus.
- Curva ascendente confirma que el servicio está activo y consumiendo CPU.

2,Panel 2 (abajo):

- Metrica prometheus_engine_query_duration_seconds_sum → indica el **tiempo total empleado en ejecutar consultas** de Prometheus.

Conclusiones

- La infraestructura híbrida creada permitió realizar pruebas reales de conectividad, monitoreo y análisis de sistemas virtualizados y contenerizados.
- Fedora Central operó exitosamente como nodo observador central, ejecutando comandos de diagnóstico, escaneo y captura de tráfico.
- Las máquinas virtuales mostraron mayor uso de recursos y complejidad, pero ofrecieron robustez en servicios de red y análisis de tráfico.
- Las herramientas Prometheus y Grafana facilitaron la recolección y visualización de métricas, proporcionando un análisis visual efectivo.
- No se pudo lograr hacer el desarrollo de arquitectura de Zabbix, se priorizo las otras dos
- El proyecto cumple con todos los criterios del segundo corte, integrando conceptos de redes, monitoreo, virtualización
- Analizar servicios y procesos mediante herramientas como htop, ps aux, glances, netstat, bpytop y journalctl.

Bibliografía

- Instalación de docker que se siguió:
<https://docs.google.com/document/d/1mo9S-6Ordr58Ukz0O7wp5cUJgUGOxv-8x1kZSqsHnUc/edit?usp=sharing>
- Archivo que nos ayudó como guia para uso de comandos y demas de las diversas MV:
<https://docs.google.com/document/d/1mo9S-6Ordr58Ukz0O7wp5cUJgUGOxv-8x1kZSqsHnUc/edit?usp=sharing>
- Prometheus Documentation: <https://prometheus.io/docs/>